Forecasting Transformer Load and Oil Temperature: Linking Data Patterns to Forecasting Errors

Group 8 | Ruxuan ZHOU, Junyang GU, Yutong SONG, Mingda WANG

0 Motivation and Study Overview

How do temporal structure, input choices and target-time conditions relate to forecasting performance?

Motivation

Predicting future transformer load and oil temperature is a practical time-series forecasting task [2]. However, an average error score does not show which variables are harder to predict or when large errors occur.
Using TimesNet model [1] on dataset "ETTm2", we examine how data patterns, input length and data scaling relate to forecasting errors.

Study Overview

1. Explore the structure of the training data (first round of exploratory data analysis, EDA).
2. Evaluate TimesNet and examine reported baseline results.
3. Test how sensitive TimesNet's forecasting performance is to input length and data scaling.
4. Examine how forecast errors relate to volatility and sudden jumps (second round of EDA: forecast-error analysis).

1 Data structure

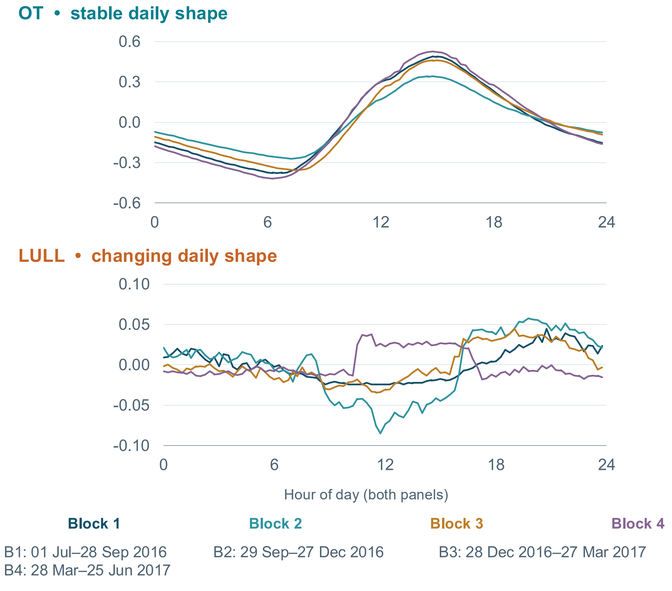
ETTm2 contains 15-minute records of transformer load and oil temperature [2]. We examine how its daily patterns and changes over time relate to forecast errors.

Data and evaluation

- 15-minute observations, seven targets: six load series and oil temperature (OT).
- Chronological split: 34,560 train / 11,520 validation / 11,520 test rows. The remaining 12,080 rows are outside this protocol [2].
- Fit external scalers on training data. Score all seven variables on a common training z-score scale.

Fig. 1 Daily shapes across periods

Training only: four consecutive 90-day blocks. Remove each day's mean, average by time of day, then divide by that variable's training SD.



B1: 01 Jul–28 Sep 2016
B2: 29 Sep–27 Dec 2016
B3: 28 Dec 2016–27 Mar 2017
B4: 28 Mar–25 Jun 2017

Minimum correlation between block profiles (six pairs per variable):
- **OT (0.985):** Highly similar daily shapes across all four periods.
- **LULL (−0.665):** Two periods show partly reversed daily shapes.

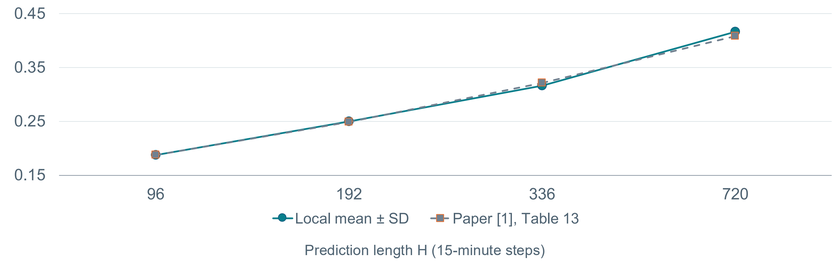
These are representative shape cases. The experiments retain all seven targets.

2 Forecasting experiments

We use TimesNet [1] as our main model, with autoregressive integrated moving average (ARIMA), XGBoost [3] and InformerLite as reference models. TimesNet identifies key periods using the fast Fourier transform (FFT), reshapes the series into two-dimensional patterns, and applies multiscale Inception convolutions. We verified 21 TimesNet runs: seven settings × three seeds (2021, 2026, 2027).

Fig. 2 TimesNet and the paper reference

We forecast all seven targets using external z-score scaling and an input length of 96 (L = 96). For our runs, points show the mean across three seeds, with error bars indicating the sample standard deviation. We report mean squared error (MSE) on the common training z-score scale.



H	96	192	336	720
Local MAE	0.267	0.306	0.347	0.404

Local MSE is within 2.12% of Table 13, but this is a scope check, not an exact replication: a later implementation with patches, 1-epoch budgets at H = 192/720, and smaller width at H = 720.

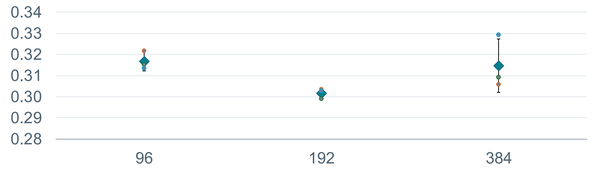
Reported baseline test summaries

Model	n	MSE 96	MSE 192	MSE 336	MSE 720	MAE 336
ARIMA	1	0.926	1.454	1.820	2.166	0.933
XGBoost	3	0.543	0.706	0.822	0.962	0.552
InformerLite	3	0.830	1.035	1.257	2.044	0.934

Fig. 3 Input length

Grey points: individual seeds. Teal diamonds: mean ± sample SD (n = 3). Both panels score MSE on the same training z-score scale.

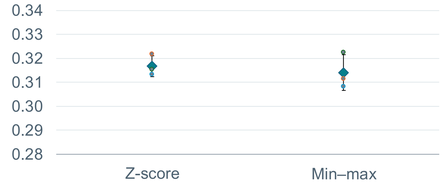
Fixed H = 336, external z-score



Using L = 96 as the baseline, L = 192 reduced MSE by 4.77%.
At L = 384, MAE rose 2.71%, and MSE's SD across seeds was 2.87 times as high.
At this longer input length, MSE rose 14.46% for LUFL but fell 11.58% for MUFL.

Fig. 4 Scaling

Fixed L = 96, H = 336



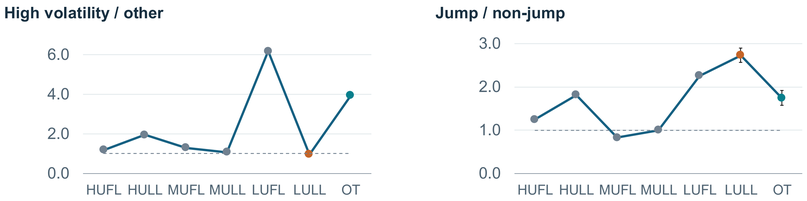
Min–max lowered MSE by 0.84% versus z-score, but this small gap does not prove equal performance. TimesNet's internal normalization remained enabled in both settings.

3 Behind the average score

Do the same variables become harder to forecast during volatile or abrupt target-time conditions?

Fig. 5 Error depends on the group

We evaluate TimesNet with L = 96, H = 336 and external z-score scaling.
Ratios are high-volatility MSE / other-time MSE (left) and jump MSE / non-jump MSE (right).
Points show the mean ratio across three seeds, with error bars indicating the sample SD.



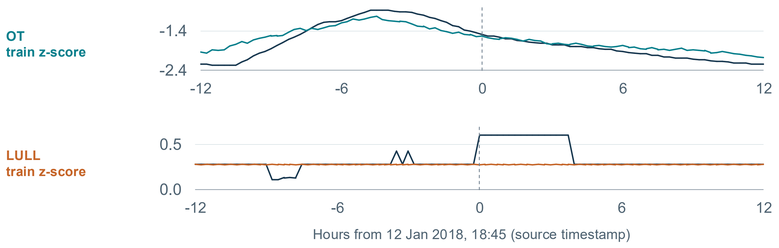
Ratios above 1 mean higher MSE during high volatility or jumps than at other times.
During high volatility, OT's MSE was 3.95× that at other times, while LULL's was slightly lower (0.96×).
At jumps, LULL's MSE was 2.73× its non-jump level, while MUFL's was lower (0.83×).

Target	High vol.	Other	Jump	Non-jump
HUFL	290,347	3,467,813	52,710	3,705,450
HULL	528,510	3,229,650	251,068	3,507,092
MUFL	374,150	3,384,010	177,004	3,581,156
MULL	220,160	3,538,000	104,742	3,653,418
LUFL	83,664	3,674,496	71,276	3,686,884
LULL	72,576	3,685,584	10,274	3,747,886
OT	31,584	3,726,576	9,111	3,749,049

Counts are forecast pairs per seed, including repeated targets. High volatility: SD of the latest 96 first differences (ddof = 1) exceeds the training 95th percentile. Jump: |Δx − training median| / robust scale > 3.5.
OT/LULL volatility cutoffs: 0.5282 / 0.7053 in recorded units. Robust scales: 0.3254 / 0.4392 (LULL uses SD fallback). Labels use observed target-time data and are retrospective.

Fig. 6 One event, the same two variables

Observed (navy) • forecast (OT teal / LULL orange)



We show rolling 24-hour-ahead forecasts (seed 2021, L = 96, H = 336) around the largest observed LULL jump with a full ±12 h window.TimesNet misses the level shift. Selection does not use prediction errors, and this single case cannot establish typical performance.

Conclusions and Limitations

Supported: moderate input extension helps this setting. Longer input does not improve every metric; scaling shows only a small difference.

Back to Fig. 1: stable OT still has high volatility-related error. Changing LULL shapes do not imply higher error in every group. These associations do not establish causality.

Limits: overlapping forecast pairs are dependent; n = 3 is small. No significance claim.

References and Resources

[1] Wu, H., Hu, T., Liu, Y., Zhou, H., Wang, J., & Long, M. (2023). TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis. International Conference on Learning Representations. https://arxiv.org/abs/2210.02186

[2] Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., & Zhang, W. (2021). Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. Proceedings of the AAAI Conference on Artificial Intelligence, 35(12), 11106–11115.

[3] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794. https://doi.org/10.1145/2939672.2939785

Data: https://github.com/zhouhaoyi/ETDataset

Implementation: https://github.com/thuml/Time-Series-Library